

Packet Tracer – Basic Wireless LAN Setup (SSID, WPA2 Security, DHCP, and Wireless Connectivity)

Objectives

Part 1: Build a Basic Wireless LAN

Part 2: Configure a Wireless Router

Part 3: Configure SSID and WPA2 Security

Part 4: Connect Wireless Clients

Part 5: Verify DHCP and Wireless Connectivity

Background / Scenario

You are setting up a small office wireless network that allows users to connect laptops and wireless devices securely.

In this lab, you will configure a wireless router, create a wireless network name (SSID), secure the network using WPA2-PSK encryption, and connect wireless clients.

Unlike previous labs that required static IP addressing, this wireless network will automatically assign IP addresses through DHCP.

This lab focuses on:

- Wireless LAN fundamentals
- SSID configuration
- WPA2 wireless security
- DHCP address assignment
- Wireless client connectivity
- Testing network communication

By the end of the lab, both wireless PCs should automatically obtain IP addresses and successfully communicate with each other.

Instructions

Part 1: Build the Wireless Network

Step 1: Add Devices

Open Cisco Packet Tracer.

Add the following devices:

- 1 Wireless Router (WRT300N)
- 2 Laptops

Rename them:

- WirelessRouter
- Laptop-A
- Laptop-B

Step 2: Verify Wireless Capability

Click **Laptop-A**.

Navigate to:

Physical Tab

Verify that a wireless NIC is installed.

If not:

1. Power off the laptop.
2. Remove FastEthernet module.
3. Install WPC300N wireless module.
4. Power on.

Repeat for Laptop-B.

Part 2: Configure the Wireless Router

Step 1: Configure LAN Address

Click WirelessRouter.

Navigate to:

GUI > Setup

Configure:

Setting	Value
Local IP Address	192.168.1.1
Subnet Mask	255.255.255.0

Step 2: Configure DHCP

Still under Setup:

Enable DHCP.

Configure:

Setting	Value
DHCP Server	Enabled
Starting IP	192.168.1.100
Maximum Users	50
Default Gateway	192.168.1.1
DNS Server	8.8.8.8

Click **Save Settings**.

Why Use DHCP?

DHCP automatically provides:

- IP Address
- Subnet Mask
- Default Gateway

- DNS Server

Without DHCP, every device must be configured manually.

Part 3: Configure Wireless Settings

Step 1: Configure SSID

Navigate to:

Wireless > Basic Wireless Settings

Configure:

Setting	Value
SSID	OfficeWiFi
Network Mode	Mixed

Click **Save Settings**.

Step 2: Configure WPA2 Security

Navigate to:

Wireless > Wireless Security

Configure:

Setting	Value
Security Mode	WPA2 Personal
Encryption	AES
Passphrase	Cisco12345

Click **Save Settings**.

Why WPA2?

WPA2 provides:

- Strong encryption
- Authentication
- Protection against unauthorized access

It is significantly more secure than WEP.

Part 4: Connect Wireless Clients

Step 1: Connect Laptop-A

Click Laptop-A.

Navigate to:

Desktop > PC Wireless

Click:

Connect

Select:

OfficeWiFi

Click Connect.

Enter:

Cisco12345

Click Connect.

Step 2: Verify DHCP Address

Navigate to:

Desktop > IP Configuration

Select:

DHCP

Expected:

Setting	Example
IP Address	192.168.1.101
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

IP address vary check IP configuration

Step 3: Connect Laptop-B

Repeat the same process.

Expected:

Setting	Example
IP Address	192.168.1.102
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

IP address vary check IP configuration

Part 5: Verify Connectivity

Step 1: Ping Wireless Router

On Laptop-A:

```
ping 192.168.1.1
```

Expected:

Reply from 192.168.1.1

4 successful replies.

Step 2: Ping Laptop-B

From Laptop-A:

ping 192.168.1.102 (can vary check ip configuration)

Expected:

Reply from 192.168.1.101

4 successful replies.

Step 3: Ping Laptop-A from Laptop-B

From Laptop-B:

ping 192.168.1.101 (can vary check ip configuration)

Expected:

Reply from 192.168.1.100

4 successful replies.

Step 4: Verify DHCP Information

On either laptop:

ipconfig

Expected:

IP Address 192.168.1.x

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

Troubleshooting Tips

If the wireless connection fails:

✓ Verify SSID spelling matches exactly

✓ Verify WPA2 passphrase matches

- ✓ Ensure DHCP is enabled
 - ✓ Ensure wireless NIC is installed
 - ✓ Verify laptops are associated with the correct SSID
 - ✓ Use:
ipconfig
to verify IP settings
 - ✓ Renew DHCP lease by selecting DHCP again
 - ✓ Verify wireless signal is connected
-

What You Learned

- How wireless LANs operate
 - How SSIDs identify wireless networks
 - How WPA2 secures wireless communication
 - How DHCP automatically assigns IP information
 - How wireless clients join a network
 - How to troubleshoot wireless connectivity
 - How to verify network communication using ping
-

Challenge Extensions (Optional)

Challenge 1: Create a Guest Wireless Network

Change SSID to:

GuestWiFi

Connect both laptops to the new network.

Challenge 2: Disable DHCP

Disable DHCP on the wireless router.

Manually configure:

Laptop-A

IP Address: 192.168.1.10

Mask: 255.255.255.0

Gateway: 192.168.1.1

Laptop-B

IP Address: 192.168.1.11

Mask: 255.255.255.0

Gateway: 192.168.1.1

Verify connectivity.

Challenge 3: Test Wrong Password Behavior

Attempt to connect using:

WrongPassword123

Observe that the client cannot join the wireless network.

Challenge 4: Add a Wired PC

Add a PC connected to the wireless router using a Straight-Through cable.

Configure DHCP.

Verify:

Wireless Client ↔ Wired PC

communication.

Challenge 5: Wireless Security Analysis

Compare:

- Open Network

- WEP
- WPA
- WPA2

Identify:

- Which is most secure?
- Which should never be used in production?
- Why WPA2 is the preferred option for small office deployments?